

Khurshid Usmanov

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Education

The University of British Columbia (UBC)

Sep 2024 – Present

Master of Science (MSc) in Physics

Vancouver, Canada

- Member of the ATLAS experiment group, working on the analysis: “Search for long-lived particles decaying into muon 2 pairs with the ATLAS detector”. **Supervisors:** Katherine Pachal, Colin Gay
- **Cumulative GPA:** 92.2/100%, (4.0/A+)
- **Highlighted Course Work:** Quantum Field Theory (I and II), Graduate Quantum Mechanics, Physics and Engineering of Particle Accelerators

University of Waterloo

Sep 2019 – Apr 2024

Bachelor of Science (BSc) in Honours Mathematical Physics (Co-op)

Waterloo, Canada

- **Minors:** Astrophysics, Pure Mathematics
- **Cumulative GPA:** 90.16/100%, (4.0/A+)
- **Undergraduate Thesis:** “Studies for the Future Muon Collider: Jet Reconstruction and Search for New Physics”, **Supervisor:** Maximillian Swiatloski (TRIUMF)
- **Highlighted Course Work:** Abstract Algebra (Group theory, Ring theory), Complex and Real Analysis, Quantum Theory (I, II, and III), Quantum Information Theory, Phase space Classical and Quantum Mechanics, Differential Geometry, Ordinary and Partial Differential Equations, Probability Theory and Statistics, Logic, Statistical Mechanics, General Relativity and Cosmology.

Research Experience

ATLAS Collaboration, UBC/TRIUMF/CERN

May 2024 – Present

Graduate Research Assistant

Vancouver, Canada

- Member of the ATLAS Long-Lived and Unconventional Physics Searches group, working on the analysis: Search for long-lived particles decaying into muon 2 pairs with the ATLAS detector
- Conducted studies of muon reconstruction efficiencies in the inner region of the ATLAS detector
- Improved and studied primary vertex reconstruction on the simulations and the LHC data
- Contributing author on a manuscript “Search for long-lived particles decaying into muon 2 pairs with the ATLAS detector”(in preparation).

ATLAS Collaboration, CERN

Jun 2025 – Jul 2025

Visiting MSc Student

Geneva, Switzerland

- Working on the analysis search for long-lived particles decaying into muon 2 pairs with the ATLAS detector
- Collaborated with the analysis lead and Phd students
- Attended CERN Research Colloquia and participated in software trainings

Muon Collider Collaboration, TRIUMF

May 2023 – Aug 2023

Undergraduate Research Assistant

Vancouver, Canada

- Developed techniques to suppress contamination from muon decays in the calorimeter detector systems
- Studied the impact on Higgs self-interaction sensitivity
- Tested boosted-jet reconstruction techniques

LoLX & nEXO Collaborations, TRIUMF

Sep 2022 – Dec 2022

Undergraduate Research Assistant

Vancouver, Canada

- Contributed to analysis software; improved waveform pulse-finding/fitting (SiPMs, PMTs)
- Participated in hardware assembly of the 2nd-generation Light Only Liquid Xenon Experiment (LoLX) detector

Institute for Quantum Computing (IQC), University of Waterloo

May 2022 – Aug 2022

Undergraduate Research Assistant

Waterloo, Canada

- Worked on a research project on achieving control in open quantum systems using Average Hamiltonian Theory
- Wrote simulations and carried out calculations in Mathematica

DarkSide-20k & DEAP-3600 Collaborations, SNOLAB

Sep 2021 – Apr 2022

Undergraduate Research Assistant

Lively, Canada

- Used GEANT4 software to carry out Monte Carlo simulations of alpha particles arising from ESR film in DarkSide-20k detector, analyzed alpha particle energy distribution and wrote a toy Monte-Carlo Simulation to describe the alpha emission
- Used DS-20k collaboration software package Pyreco to simulate the response in the TPC of DS-20k
- Conducted a study of time fitting of the signal response of DarkSide-20k detector due to alpha particle interaction with liquid argon (Ar-40) in the DS-20k vessel to reject accidental surface events
- Created plots, visualizations and summarized the results of the study in a technical report stored in DarkSide-20k Database
- Assisted with hardware installations for the DEAP-3600 detector at the SNOLAB underground facility

Department of Applied Mathematics, University of Waterloo

Jan 2021 – Apr 2021

Undergraduate Research Assistant

Waterloo, Canada

- Modeling and numerical tools for flexible filament dynamics; MATLAB simulations of bacterial swimming.
- Created models and numerical tools for simulating the dynamics of flexible filaments in viscous fluids in MATLAB
- Developed and explored simulations of swimming bacteria in MATLAB using a newly derived approximation method to determine the average velocities of bacteria

Programs, Conferences & Presentations

Tri-Institute Summer School on Elementary Particles (TRISEP)

Jun 2025

TRIUMF, Perimeter Institute, SNOLAB

Vancouver, Canada

ATLAS Canada Week

May 2025

Simon Fraser University

Vancouver, Canada

Winter Nuclear and Particle Physics Conference (WNPPC)

Feb 2025

Banff, Canada

- Talk: “Efficiency study of the displaced muon reconstruction for dark photon search with ATLAS detector.”

Canadian Undergraduate Physics Conference

Oct 2023

Waterloo, Canada

- Talk: “Studies for the future Muon Collider: Jet Reconstruction and Search for New Physics.”

Perimeter Scholars International (PSI) Start Program

Remote

Perimeter Institute for Theoretical Physics

May–Jul 2023

Canadian Undergraduate Physics Conference

Oct 2022

Guelph, Canada

- Talk: “Study of the time structure of S1 Signals from the Surface of DarkSide-20k.”

Undergraduate School on Experimental Quantum Information

May–Jun 2021

Institute for Quantum Computing (IQC), University of Waterloo

Waterloo, Canada

Canadian Astroparticle Physics Summer School

May 2021

Queen’s University, SNOLAB

Kingston, Canada

Awards & Scholarships

- International Graduate Tuition Award, UBC (Aug 2025) – 3,200 CAD
- Faculty of Science Master’s Tuition Award, UBC (Aug 2024) – 5,400 CAD
- International Graduate Tuition Award, UBC (Aug 2024) – 3,200 CAD
- I.R. Dagg Memorial Scholarship, University of Waterloo (Dec 2023) – 1,000 CAD
- Undergraduate Research Award, Institute for Quantum Computing (Feb 2022) – 10,000 CAD
- President’s Scholarship, University of Waterloo (Aug 2019) – 2,000 CAD
- International Student Entrance Scholarship, University of Waterloo (Aug 2019) – 15,000 CAD

Teaching Experience

Teaching Assistant at the University of British Columbia

Aug 2024 - Present
Winter 2026

- PHYS 304: *Quantum Mechanics*
- PHYS 200: *Introduction to Special Relativity and Quantum Mechanics*
- PHYS 306: *Advanced Classical Mechanics*
- PHYS 200: *Introduction to Special Relativity and Quantum Mechanics*

Fall 2025
Winter 2025
Fall 2024

Outreach Activities

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- **Lecture:** “Particle Physics at the LHC: Searching for Dark Photons with the ATLAS Detector” as a part of the Space Camp event series at H.R. MacMillan Space Centre *June 2026*
 - **Short Talk:** “Following Muons into the Dark: ATLAS Experiment and the Search for Hidden Particles” as a part of the Nerd Nite North Vancouver event series *May 2026*
 - **Lecture:** “Following Muons into the Dark: ATLAS Experiment and the Search for Hidden Particles” as a part of UBC Physics Circle Initiative organized by the UBC Department of Physics and Astronomy *Dec 2025*
 - **Lecture:** “The Path to Particle Physics: From Underground Labs to CERN” to a community of entrepreneurs in Tashkent, Uzbekistan *Aug 2025*
 - **Lecture:** “Career and pathways in STEM. From an Undergraduate Student to to a Research Associate, Intro to Particle Physics.” to the general public at GroundZero coworking center in Tashkent, Uzbekistan *Aug 2024*
 - **Lecture:** “Career and pathways in STEM. From an Undergraduate Student to to a Research Associate. Intro to Particle Physics” for the high school students at Tashkent International School, Tashkent, Uzbekistan *Aug 2024*
 - **Lecture:** “The pathway of a young physicist: from researching dark matter 2000m underground to working at the particle accelerator complex” to the general public at GroundZero coworking center in Tashkent, Uzbekistan *May 2023*
 - **Lecture:** “Physics of Quantum Information and Emerging Technologies” to the general public at GroundZero coworking center in Tashkent, Uzbekistan *Aug 2022*

Leadership & Service

Member, the Graduate Students and Postdocs Society (GAPS) TRIUMF

Sep 2025 – Present
Vancouver, Canada

- Organize and manage events for TRIUMF’s graduate and postdoctoral community

Steward, Graduate Teaching Assistants Union CUPE 2278, University of British Columbia

Sep 2025 – Present
Vancouver, Canada

- Serve as TA representative for the Department of Physics and Astronomy, counseling TAs on policy, work issues, and mediation strategies.

Student Representative

Jan 2023 – Apr 2024
Waterloo, Canada

Department of Physics and Astronomy, University of Waterloo

- Represented fellow Mathematical Physics students in the Department of Physics and Astronomy
- Met with faculty each term to provide feedback on physics courses and labs
- Obtained feedback from fellow physics students and informed the Department

Mathematical Physics Program Representative & Team Leader Science Ambassador Program, University of Waterloo

Aug 2020 – Apr 2024
Waterloo, Canada

- Represented the faculty and the university at recruitment events throughout the year, promoted and addressed the questions about Waterloo Science to encourage applicants to join the university
- Conducted in-person campus tours at the UWaterloo for prospective students and their guardians
- Consulted prospective students at virtual meetings and guided them through the admission process for the

UWaterloo Physics programs

Vice President

Jan 2023 – Jan 2024

University of Waterloo Undergraduate Physics Society (PhysClub)

Waterloo, Canada

- Assisted in the day-to-day operations of the club and was responsible for the conduct of the executive staff and officers.
- Represented the club at community meetings, including faculty meetings and meetings with the Science Society Board of Directors
- Organized and oversaw educational and social events for the physics undergraduate students' community at UWaterloo

External Affairs Executive

Nov 2022 – Nov 2023

Canadian Undergraduate Physics Conference (CUPC)

Waterloo, Canada

- Communication with external parties: sponsors, speakers, venue, caterers
- Event planning and organization of lab and building tours of the Institute for Quantum Computing (IQC) and the Perimeter Institute for Theoretical Physics

Physics Seminar Organizing Team

Aug 2022 – Jan 2023

Department of Physics and Astronomy, University of Waterloo

Waterloo, Canada

- Organized invited talks and physics film events; outreach to prospective speakers; logistics support.
- Brought together physics students in all years to hear invited speakers, view physics-related films, and learn about current research
- Contacted potential speakers within the Department of Physics and Astronomy at UWaterloo and outside, assisting with seminar organization and logistics

Technical Skills

- **Python** (NumPy, Pandas, SciPy, Matplotlib, Awkward Array, Uproot)
- **Bash Scripting**
- **C++**
- **Jupyter Notebooks**

Languages

English: Advanced **French:** Advanced (DELF B2) **Russian:** Native **Uzbek:** Native

References

- **Katherine Pachal** – Research Scientist, Adjunct Professor, TRIUMF, the University of British Columbia (kpachal@triumf.ca [🔗](#))
- **Colin Gay** – Professor, The University of British Columbia (cgay@phas.ubc.ca [🔗](#))
- **Maximilian Swiatlowski** – Research Scientist, TRIUMF (mswiatlowski@triumf.ca [🔗](#))
- **Chris Jillings** – Research Scientist, SNOLAB (chris.jillings@snolab.ca [🔗](#))
- **Richard Epp** – Associate Professor, Undergraduate Advisor for Mathematical Physics, University of Waterloo (rjepp@uwaterloo.ca [🔗](#))